

Experiment 6: Techniques for Isolation of Pure Cultures

Principle:

Microorganisms exist in nature as mixed populations. However, to study and microorganisms in the laboratory, we must have them in the form of a **pure culture**; that is, one in which all organisms are descendants of the same organism.

Cultures on solid media appear as visible mass of growth, called **colonies**. A **colony** forms on a plate when a single microbe is inoculated onto the surface of the plate and reproduces until there are enough cells to form a visible mass. Since a colony is formed from a single cell, a colony should then represent a pure culture.

Pure cultures (colony) can be isolated on solid media using streak plate, pour plate and spread plate methods. A streak plate provides a rapid method of diluting the sample by streaking many lines across the agar surface with inoculating loop. Spread plate and pour plate are used as method of enumerating bacteria. A series of dilutions are always prepared, and a small volume from the diluted bacterial mixture is taken for spread or poured to obtain separate and countable colonies. Dilutions are usually performed in series of 1/10 dilutions (figure 6). You will learn about serial dilution in the next experiments.

Objective:

- Isolate bacteria in pure culture from a specimen containing mixed populations
- Practice streak plate, pour plate and spread plate for isolation of pure colony
- Understand the purpose of streak plate, pour plate and spread plate techniques

Technique 1: Streak Plate Method

The streak plate allows progressive dilution of inoculates of bacteria over the surface of agar in such a way that colonies grow well separated from each other. The purpose of streaking a culture on a plate is to dilute the culture enough to get isolated colonies. Isolated colonies are needed to define colonial morphologies and detect different biochemical characteristics. In addition, streak plating is performed to purify bacterial cultures before further analysis is performed. Because bacteria often exist as clumps or chains, stringent colony purification involves streaking isolated

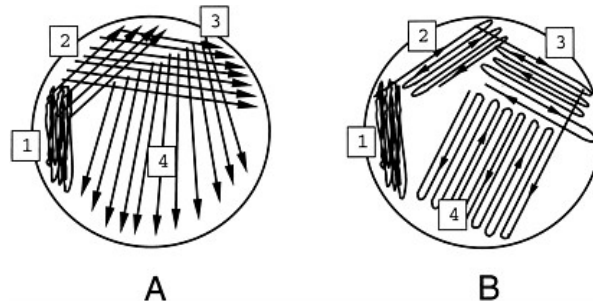
colonies at least twice: first to isolate a colony, then a second time to assure that the clump or chain that started the initial colony was homogeneous. There are many different streaking patterns and as long as isolated colonies are achieved, no methodology is wrong.

Materials and Equipments

Specimen (mixed broth culture)	Nutrient agar plate
Inoculating loop	Bunsen burner
Incubator	

Procedure

1. Divide the agar plate in to four quadrants
2. Flame the inoculating loop red hot and allow it to cool.
3. If streaking from broth, open the tube of broth, flame the neck of the tube and place the loop into the broth.
4. If streaking from a spread plate colony, touch the loop to the center of an isolated colony.
5. Partially lift the lid of the Petri dish containing the sterile agar, just enough to get the loop in
6. Hold the loaded loop parallel with the surface of the agar; streak over the agar surface starting from one edge across a small area of the medium (**area 1 in figure 5**)
7. Flame the loop and allow it to cool. Turn the dish through 90° anticlockwise. Make the next streak lines across the initial streak. Make sure that a small amount of culture is carried over (**area 2 in figure 5**)
8. Flame the loop and allow it to cool. Turn the dish through 90° anticlockwise again and streak across the second streak into area 3 (figure 5).
9. Flame the loop and allow it to cool. Turn the dish through 90° anticlockwise and streak the fourth set of lines into the centre (**area 4 in figure 5**). Use fewer cross-streaks here than in the previous quadrant
10. Remove the loop and close the Petri dish; reheat the loop before placing to its racks
11. Incubate the plate in an inverted position at 37°C for 24 hr.



Parallel line quadrant streak

undulating line quadrant streak

Figure 1 Steps involved in streaking a plate for isolated colonies

Technique 2: Pour Plate Technique

A pour plate: is one in which a measured volume (usually 1 ml) of a diluted sample is added to sterile petri plate and mixed with a molten cooled agar medium (48 °C) until the bacteria are evenly distributed throughout the liquid, and then allowed to solidify trapping the bacteria at separate discrete positions within the matrix of the medium. The growth will probably have distinct colonies that can be picked for further purification. Pour plates allow microorganisms to grow both on the surface and within the medium. Most of the colonies grow within the medium and are small in size; the few that grow on the surface will be large in size. If the dilution and volume are known, the number of viable bacteria per ml of the food sample can be determined. The main disadvantages of the Pour Plate method are that the bacteria must be able to withstand the 48 °C agar for several minutes, you must work fast to ensure that the agar does not solidify before you can pour the plates, and it requires facilities for keeping the agar molten.

Materials and Equipments:

Specimen (mixed broth culture)	Nutrient agar pour bottle at 48 °C
Petri plates	Test tubes
Peptone water diluents	Sterile pipettes
Bunsen burner	Incubator

Procedure:

1. Label dilution tubes and petri plates with the respective dilutions to be plated
2. Prepare serial dilution of sample (10^{-1} - 10^{-3}), and from each dilution transfer 1 ml to respectively labeled petri plate using separate pipette

3. Pour molten agar, tempered to 48-50°C, into each Petri dish, and close it.
4. Mix the plate well by gently moving in vertical, clockwise, horizontal and anticlockwise direction as shown in the figure below, on the table top. **Note that** the agar must not touch the lid of the plate and the surface must be smooth with no bubbles
5. Let the agar to set.
6. Incubate the plates in an inverted position at 37°C for 24 hr.
7. Observe and characterize colonies within and on the surface of the medium

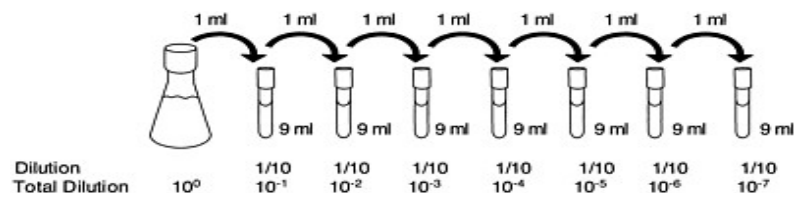


Figure 2 Simple serial dilution series using 9 ml blanks

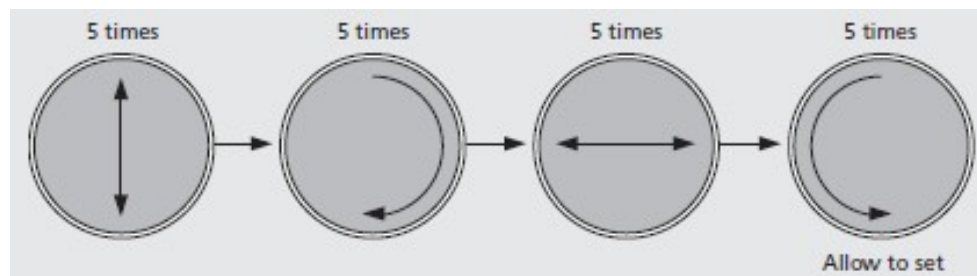


Figure 3 Mixing the pour plates

Technique 3: Spread plate technique

Spread plate: a small volume of diluted sample (usually 0.1ml) is evenly spread onto the surface of solidified agar plate with a sterile bent-glass rod. The glass rod is normally sterilized by dipping in alcohol and flamed to burn off. It may also be sterilized in a hot air oven wrapped with aluminum foil. It is advisable to use agar plates that have a well-dried surface so that the inoculum dries quickly. Dry the surface of agar plates by either incubating the plates for several hours, with the two halves separated and the inner surfaces directed downwards. At the end of incubation bacterial growth is visible as surface colonies. The bacterial count can be determined, as well as this can be used to test antimicrobial sensitivity of the growing bacteria.

Materials:

Nutrient agar plate

L-shaped glass rod (spreader)

Pipette

Ethanol

Incubator

Procedure:

1. Label petri plates with names and dilutions to be plated
2. Transfer 0.1 ml of the diluted sample as above onto surface of labeled dried agar plates
3. Dip a glass spreader into alcohol, flame and let the alcohol burn off, and allow cooling.
4. Place the spreader onto the inoculated agar, rotate the plate with left hand and move the spreader in a top-to-bottom or a side-to-side motion to spread the inoculum over the surface of the agar. Make sure the entire agar surface is covered.
5. Replace the lid of the Petri dish. Flame spreader with alcohol
6. Let the inoculum dry; incubate the plate in the inverted position at 37 °C for 24 hr.
7. Observe and characterize colonies on the surface of the medium

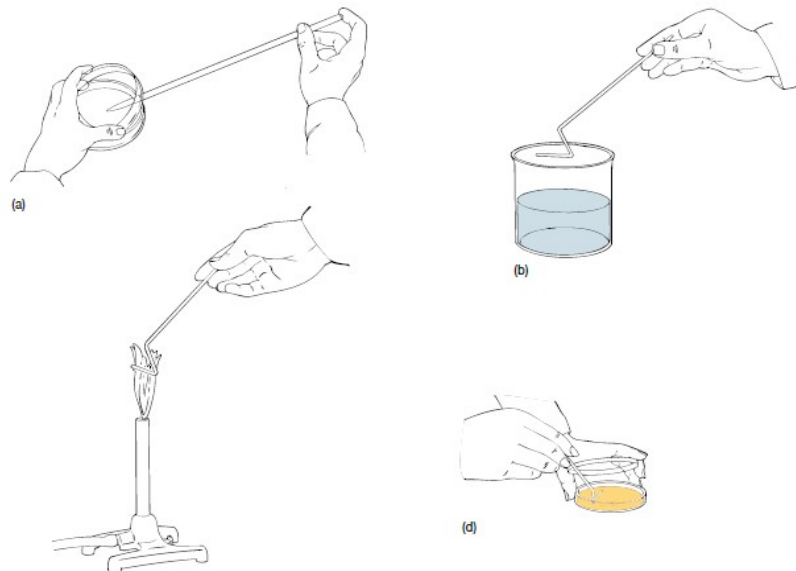
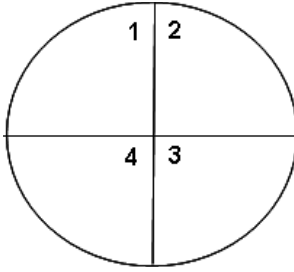


Figure 4 Spread plate technique

Discussion Questions:

1. Draw the quadrant streak method on the plate. Which area of a streak plate will contain the greatest amount of growth? The least amount of growth? Explain your answers.



2. Did your colonies on the streak plate appear homogeneous after streaking? Using techniques from this lab, how could you further purify these cultures, and how would you confirm their purity?
3. Why is it important to start a bacterial culture with a single, isolated colony?
4. What is a bacterial colony?
5. What is the purpose of the ethanol in the spread-plate technique?
6. What is the purpose of the spread-plate technique?
7. How would you determine whether culture media given to you are sterile before you use them?